# Matthew Hockert

PhD Student in Applied Economics · University of Minnesota

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## Education

*PhD in Applied Economics* September 2022 – Present
University of Minnesota, Minneapolis, MN

- Fourth-year student (2025–2026). Advisor: Elizabeth Davis.
- Fields: Labor and Population Economics, Policy Analysis.
- Research Fields: Land Use, Housing, Transportation, Public Finance.
- Methods: Spatial econometrics, causal inference (DiD, RD, IV), forecasting.

*Master of Science in Applied Economics* September 2020 – June 2024
University of Minnesota, Minneapolis, MN

- Graduate minor in Data Science.
  - Courses: Machine Learning Fundamentals, Principles of Database Systems, Theory of Statistics.
- Thesis: "A Spatial Approach to Agrifood Supply Chain Structure" (Advisor: Hikaru Peterson).
- Recipient of the UMN Applied Economics MS Thesis Award, 2024–25.
- Selected Paper Presenter at the Agricultural & Applied Economics Association (AAEA) Annual Meeting, 2023.

*B.A. in Economics and Political Science* 2017 – 2020
University of Minnesota Duluth

- GPA: 3.7/4.0. *Cum Laude*. University Honors Program, Dean's List.

## Works in Progress

*Dissertation Papers*

- "Conservation and Externalities: The Effect of Conservation Easements on Neighboring Residential Parcels"
  - Staggered DiD on 550K residential parcels; estimates property tax and value spillovers from nearby easements.
- "Transportation Changes Land Use: How the St. Croix Bridge Shaped Accessibility, Housing Values, and Travel Behavior"
  - IV-DiD exploiting a bridge opening; decomposes accessibility capitalization into land vs. improvement values.
- "Rent Control Capitalization into Land Values: Evidence from New Jersey's New Construction Exemption"
  - Triple-difference design on 2024 assessed values exploiting New Jersey's New Construction Exemption (rent-control exemption for new 4+ unit buildings); finds a 54% land value premium, concentrated in land with no effect on structures.
- "When Do Local Income Taxes Cause Residential Sorting? Evidence from Pennsylvania's Piecewise Earned Income Tax"
  - Continuous stacked difference-in-differences on LEHD/LODES worker flows (2002–2019) exploiting the piecewise "max-rule" of Pennsylvania's local Earned Income Tax; a binding decomposition isolates rate changes that move a worker's bill and finds residential sorting away from binding municipal increases.

*Additional Research*

- "Restrictive Rules, Restrained Housing: Rent Control Policies and Their Local Impacts"
  - Developed LLM-based algorithm to extract and classify rent control ordinances across 35 variables from municipal codes spanning 111 New Jersey jurisdictions.

## Research Experience

*Conservation Economist – Intern II* June 2024 – Present
Hennepin County, Minneapolis, MN

- Evaluated the impact of conservation easements and programs (Green Acres, Agricultural Preserve, Rural Preserve) on property values, tax burden, and neighboring parcels.
- Studied spillover effects on neighboring parcels, including shifts in housing values and tax liability from nearby conservation easements.
- Simulated conservation easement scenarios to estimate fiscal impacts and tax implications for municipal budgets.
- Used R, QGIS, and ArcGIS to analyze spatial patterns in conservation and tax policy effects on property values.
- Administered surveys and managed data using Survey123.

*Graduate Research Assistant* September 2022 – Present
Accessibility Observatory, University of Minnesota, Minneapolis, MN

- Developed Python pipeline to ingest data from AWS, clean it, and produce weighted-average job accessibility measures from block to state level across the U.S.
- Built R scripts using OpenStreetMap network data and the R5 routing engine to calculate ambulance response and transport times to Level I–III trauma centers for every highway segment in rural Minnesota.
- Created public datasets on multimodal transportation accessibility (bike, car, transit) using Pandas and SQL.
- Built additional data pipelines for ad hoc deliverables for the Metropolitan Council and state Departments of Transportation.
- Developed R Shiny application with Leaflet to display national job accessibility from block group to state level.

*Data Science Intern* June 2022 – March 2024
Federal Reserve Bank of Minneapolis, Minneapolis, MN

- Built forecasting models (VAR, Kalman Filter) for macroeconomic stress testing, assessing bank resiliency against economic shocks.
- Converted Stress Testing production code from MATLAB to R following software development best practices.
- Built R Shiny applications incorporating predictive risk models to identify bank risk levels by loan and asset type.
- Investigated changes in rural banking quality and agricultural output in the Ninth District.
- Created R Shiny dashboard using Census ACS and Federal Reserve data to compare county-level economic and banking conditions.
- Presented analysis of U.S. banking conditions under CECL policy to examiners and supervisors.

*Graduate Research Assistant* September 2020 – 2022
Foodshed – Agricultural Commodities Project, University of Minnesota, Minneapolis, MN

- Examined regional food system resiliency during the COVID-19 pandemic using OLS and spatial regression models in R.
- Built comprehensive dataset integrating Census and IMPLAN data.

*Undergraduate Research Assistant* September 2019 – May 2020
Bureau of Business and Economic Research, University of Minnesota Duluth

- Performed local economic data analysis using IMPLAN, Excel, and SPSS.
- Cleaned and interpreted survey data in Qualtrics; co-authored publications on university website.
- Contributed to literature review and public outreach for labor market research.

## Presentations

- "Transportation Changes Land Use: How the St. Croix Bridge Shaped Accessibility, Housing Values, and Travel Behavior" – Transportation Research Board (TRB) Annual Meeting, Washington D.C., January 2026.
- "Transportation Changes Land Use: How the St. Croix Bridge Shaped Accessibility, Housing Values, and Travel Behavior" – Transportation Research Conference, Center for Transportation Studies, October 2025.
- "Calculating the Total Travel Time from Crash to Emergency Care in Rural Minnesota" – Transportation Research Conference, Center for Transportation Studies, November 2024.
- "A Spatial Approach to Agrifood Supply Chain Resiliency" – Selected Paper Presenter, AAEA Annual Meeting, July 2023.

## Technical Skills

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| **Programming:** | R, Python, SQL |
| **ML & Statistics:** | Forecasting (VAR, Kalman Filter), spatial regression, panel data methods, causal inference |
| **Data & Cloud:** | AWS (S3, EC2), DBeaver, Pandas, Linux |
| **AI Tools:** | Large language models for data extraction and classification |
| **Geospatial:** | QGIS, ArcGIS |
| **Visualization:** | Tableau, R Shiny, Leaflet |
| **Version Control:** | Git, GitHub, GitLab |

## Honors and Awards

- UMN Applied Economics MS Thesis Award, 2024–25
- University Honors Program, University of Minnesota Duluth
- Dean's List (2017–2020)
- *Cum Laude*, Economics Department Honors, Omicron Delta Epsilon

## Professional Memberships

Agricultural & Applied Economics Association (AAEA)